23/12/2024 FE SEM-I ALL BRANCHES (NEP 2020) APPLIED CHEMISTRY QP CODE: 10070837

Time: 1.30 Hours **Marks: 45**

N.B.
1. **Question number 1 is compulsory.**
2. **Attempt any two questions from Q.2 to Q.5.**
3. **Draw neat diagrams and write chemical equations wherever necessary.**
4. **Figures to right indicate full marks.**

1. Solve any Five.

(a) What are the characteristics of good fuel? 3

(b) Define corrosion, how you will join the metals to prevent corrosion, justify with diagram. 3

(c) Give the purposes of making of an alloy. 3

(d) Explain the term 'glass transition temperature' with significance. 3

(e) Define Matrix phase and dispersed phase with example. 3

(f) Write are the principles of Green Chemistry. 3

(g) 0.5 gm. of coal sample was burnt in Bomb calorimeter produced 0.06 gm. of $BaSO_4$. calculate percentage of sulphur 3

2 . (a) What is cathodic protection? How is it done by using impressed current and sacrificial anode? Explain with suitable examples. 6

(b) Explain with neat suitable diagram determination of Carbon and hydrogen. by ultimate analysis 5

(c) What do you mean by green solvent, explain it with suitable example. 4

3. (a) A Polymer has the following composition: 100 Molecules of molecular mass 1000, 200 molecules of molecular mass 2000 and 500 molecules of molecular mass 5000.Calculate the number and weight average of molecular weight and the polydispersity index. (PI) 6

(b) What is condensed phase rule equation? Explain its application with the help of phase diagram to two component lead-silver system. 5

(c) Explain Laminar and sandwich panel of structural composite. 4

4. (a) Explain following properties of polymer a) Electrical b) Optical. 6

(b) Explain synthesis of Indigo dye by traditional way and by green pathway. 5

(c) A sample of coal has the following composition C = 80 %, O = 8%, H = 6 %, S = 1.5%, N = 1 %, Calculate the higher and lower calorific values of coal. 4

5. (a) Explain the effect of following factors on the rate of corrosion : 6

(1) Nature of corrosion product. (2) pH

(3) Anodic and cathodic areas (4) purity of metal

(b) Explain a) Fiber glass reinforced composite. 5

b) Carbon fiber reinforced composite.

(c) An alloy of tin and lead contain 73% tin. Find the mass of eutectic in 1kg of solid Alloy, if the eutectic contains 64% of tin. 4

FE sem - I (NEP) | Branch - All | NOV - 2025 | Q.P. code - 96621 1|1

Date - 26-12-2025

Time: 1:30 hours Marks: 45

N.B.
1. Question No. 1 is compulsory.
2. Attempt any two questions from Question No.2 to Question No.5.
3. Draw a neat and well-labelled diagram, and write the chemical reaction wherever necessary.
4. Marks are given on the right side of the questions.

1. ATTEMPT ANY FIVE QUESTIONS.

(a) Define calorific value and write the significance of proximate analysis of coal. 03

(b) Explain the sacrificial anode cathodic protection method with a neat and well-labelled diagram. 03

(c) Explain the terms involved in Gibbs phase rule with an example of a one-component system. (Phase, Component and Degree of Freedom) 03

(d) Define the tensile strength of a polymer and explain its significance. 03

(e) Discuss applications of composites. 03

(f) Discuss super critical solvents with its applications. 03

(g) How is the percentage of sulphur determined in a solid fuel? 03

2. (A) How can material selection and design prevent corrosion? Explain which protective coatings can be used for corrosion prevention. 06

(b) Calculate the gross and net calorific value of coal having the following compositions. Carbon=85%, hydrogen=8%, sulphur=1%, nitrogen=2%, ash=4%, and the latent heat of steam is 587 cal/gm. 05

(c) Write the conventional and green methods for the synthesis of adipic acid. 04

3. (a) Explain all types of chemical bonds existing in polymers. 06

(b) Write notes on (a) reduced phase rule(c) applications of phase diagram. 05

(c) Classify composites with the help of a diagram and an example of each class. 04

4. (a) Write short notes on crystallinity, number average molecular weight and fatigue of polymers. 06

(b) Write about the synthesis and advantages of biodiesel. 05

(c) 1.0 g of coal sample was digested (Kjeldahl method). The liberated ammonia was absorbed in 25 ml of N/10 H_2SO_4. To neutralise excess acid, 15 ml of 0.1 N NaOH was required. Determine the % nitrogen in the given sample of coal. 04

5. a) Explain the rusting of iron with the help of the electrochemical theory of corrosion and a diagram. 06

(b) Define matrix and dispersed phase with its functions and examples. 05

(c) An alloy of Cd and Bi contains 25% Cd. Find the mass of eutectic in 1 kg of alloy if the system contains 40% Cd. 04

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Paper / Subject Code: N10413 / Applied Chemistry

09/06/2025 FE ALL BRANCHES SEM-I (NEP-2020) AC QP CODE: 10082409

Time:1 $^{1}/_{2}$hour **Maximum marks :45**

NB:

1) Question No.1 is Compulsory
2) Attempt any two questions from the remaining four questions
3) Draw neat diagrams and write chemical equations wherever necessary.
4) Figures to the right indicate full marks.

Q. 1 Attempt any **five** of the following: (15)

a) What are advantages of gaseous fuels over solid and liquid fuels?

b) Distinguish between galvanizing and tinning.

c) Why are alloys preferred over pure metals in many applications?

d) Explain viscoelasticity property of polymer.

e) Define Matrix phase and dispersed phase of composites.

f) What is supercritical CO_2, and how does it differ from liquid and gaseous CO_2.

g) 1 gm. of coal sample was used for determination of nitrogen. The ammonia evolved was passed into 50ml of 0.1 N H_2SO_4, the excess acid required 42 ml. of 0.1N NaOH for neutralization Calculate % of N.

Q.2 a) How do following factors affects the rate of corrosion i) P^H medium ii) Purity of metal iii) Area of anode and cathode. (6)

b) Explain the determination of sulphur present in coal sample and write its significance. (5)

c) Write an informative note on biodiesel. (4)

Q.3 a) In a polymer sample 30% molecules have a molecular mass 20,000, 40% have molecular mass 30,000 and the 30% have 60,000. Calculate the weight average, number average molecular weight and polydispersity of polymer sample. (6)

b) What is condensed phase rule equation? Write merits and limitation of phase rule. (5)

c) Write a short note on sandwich panel of structural composite write its applications. (4)

Q.4 a) Give the definitions and significance of following properties of polymer. Tensile strength, insulation resistance and refractive index. (6)

b) Explain synthesis of Adipic acid by traditional and green pathway. Also write the principle of green chemistry involved in this reaction. (5)

c) A sample of coal has the following composition C = 84 %, O = 8.4 %, H = 5.5 %, S = 1.5%, N = 0.6 %, Calculate the higher and lower calorific values of coal. (4)

Q.5 a) How material selection and proper design of equipments can prevent corrosion. Explain with the help of diagram and designing principles. (6)

b) Explain fiber reinforced composite and write its applications and limitations. (5)

c) An alloy of Cd and Bi contain 25% of Cd Find the mass of eutectic in 1kg of solid Alloy, if the eutectic contains 40 % of Cd. (4)

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